Lecture 15

Monday, April 6, 2026 7:23 PM

Today: Gradient Descent (and its variants) for Linear Regression

Loss function: $L(\theta)$

↓ function ↳ parameter

$$\theta^* = \text{argmin } L(\theta)$$

Gradient : $L : \mathbb{R}^d \to \mathbb{R}$

$$\nabla_\theta L = \begin{bmatrix} \frac{\partial L}{\partial \theta_0} \\ \frac{\partial L}{\partial \theta_1} \\ \vdots \\ \frac{\partial L}{\partial \theta_d} \end{bmatrix} \in \mathbb{R}^d$$

$$L(\theta) = \theta_1^2 + 2\theta_2$$

$$\nabla_\theta L = \begin{pmatrix} 2\theta_1 \\ 2 \end{pmatrix}$$

$\theta_1 = 3$ $\theta_2 = 5$

$(3, 5)$ $(3+\delta, 5)$

$(3-\delta, 5)$

$\frac{\partial L}{\partial \theta_i}$ measure of how L increases if ~~wt~~ we nudge the i^{th} dimension keeping every other dimension constant

- Gradient points in the direction of the steepest ascent
- Moves in the direction of $-\nabla_\theta L(\theta)$

$$\theta \leftarrow \theta - \eta \nabla_\theta L(\theta)$$

- (−) : -ve
- η : step size
- $\nabla_\theta L(\theta)$: Gradient

← Gradient Descent

Linear Regression

d-1 features $X = [1 - - - - -]$ ← 1^{st} sample

$\leftarrow i^{th}$ sample
$\leftarrow n^{th}$ sample
$n \times d$ matrix

$y^{(i)}$ — value corresponding to sample i

$$L(\theta) = \underbrace{\frac{1}{2n}}_{\text{constant, added for conviences}} \|X\theta - y\|^2 = \frac{1}{2n} \|y - X\theta\|^2 \leftarrow$$

$$\theta^* = \underset{\theta}{\operatorname{argmin}}\ L(\theta)$$

- θ^* can be found using linear algebra

$$\nabla_\theta L(\theta) = \underbrace{\frac{1}{n}}_{1\times 1} \underbrace{X^T}_{d\times n} \underbrace{(X\theta - y)}_{n\times 1} = \underbrace{\frac{1}{n} X^T (\hat{y} - y)}_{d\times 1}$$

$\underbrace{(1\times 1)(d\times n)(n\times 1)}_{d\times 1}$

Proof omitted (verify) $\qquad \theta \leftarrow \theta - \eta \nabla_\theta L(\theta)$

$$\theta \leftarrow \theta - \frac{\eta}{n} X^T (X\theta - y)$$

(Batch) Gradient Descent for linear regression

- $X\theta$: $O(nd)$ multiplications
- $X\theta - y$: $O(n)$ subtractions
- $X^T(X\theta - y)$: $O(nd)$ multipcation

$\begin{bmatrix} & \\ & \\ & \end{bmatrix}_{n\times d} \begin{bmatrix} \\ \end{bmatrix}_{d\times 1}$

$\underline{O(nd)}$ computations needed for each step

Housing market : $n = 10^7$ $d = 50$: 5×10^8
Image processing : $n = 10^6$ $d = 10^5$: 10^{11}

Memory requirement is also (prohibitively) large.

Stochastic Gradient Descent

→ $L(\theta) = \frac{1}{2n} \|X\theta - y\|^2$

Core Idea

$\underbrace{L^{(i)}(\theta)} = \frac{1}{2} (\hat{y}^{(i)} - y^{(i)})^2$

contribution of the i^{th} sample to the loss

→ $\nabla_\theta L^{(i)}(\theta) = (\hat{y}^{(i)} - y^{(i)}) x^{(i)}$

$\theta \leftarrow \theta - \eta \nabla_\theta L^{(i)}(\theta) = \theta - \eta (\hat{y}^{(i)} - y^{(i)}) x^{(i)}$

① how expensive is this step — $O(d)$
② why does this make sense ?

① $\hat{y}^{(i)}$: $O(d)$ ←
$\hat{y}^{(i)} - y^{(i)}$: $O(1)$
$(\hat{y}^{(i)} - y^{(i)}) x^{(i)}$: $O(d)$ ←

Overall : $O(d)$

→ i – uniformly at random

② $\mathbb{E}[\underbrace{\nabla_\theta L^{(i)}(\theta)}] = \frac{1}{n} \sum_{i=1}^{n} \nabla_\theta L^{(i)}(\theta) = \nabla_\theta L(\theta)$

(proof omitted)

On average, stochastic gradient points in the right direction

SGD Gradient is unbiased

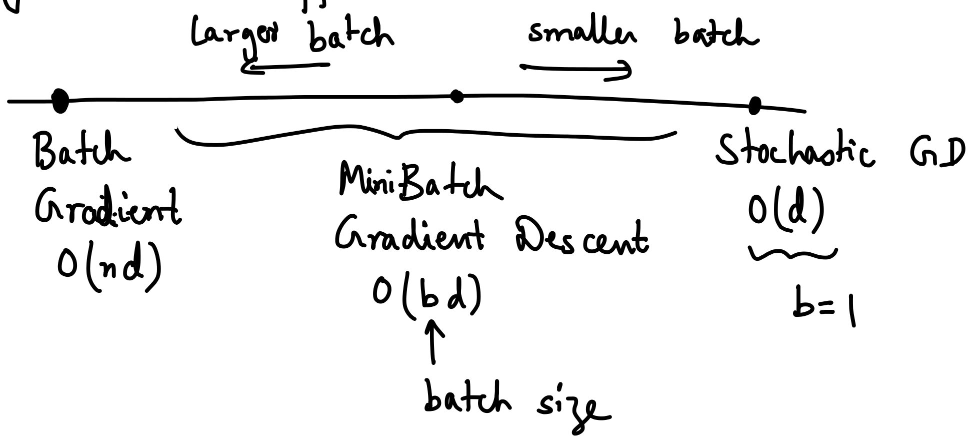
$\theta \leftarrow \theta - \underbrace{\eta_t} \nabla_\theta L^{(i)}(\theta)$

decrease with time : $\eta_t = \frac{1}{\sqrt{t}}$

SGD algorithm

1. Initialize θ
2. For each epoch (pass through all n datapoints)
   a) shuffle the data (permutation of $1:n$)
   b) for each sample $(x^{(i)}, y^{(i)})$ in the shuffled order
      Compute $\hat{y}^{(i)} = x^{(i)T}\theta$
      Update $\theta \leftarrow \theta - \eta(\hat{y}^{(i)} - y^{(i)})\, x^{(i)}$
3. Repeat until ....

Why do we shuffle?



Mini-batch Gradient Descent

Pick B data-points (uniformly at random)

$$\nabla_\theta L_B(\theta) = \frac{1}{B}\sum_{i\in B}(\hat{y}^{(i)} - y^{(i)})\, x^{(i)} = \frac{1}{B} X_B^T (X_B\theta - y_B)$$

X_B : X restricted to row corresponding to the mini-batch

$$\theta \leftarrow \theta - \frac{\eta}{B} X_B^T (X_B\theta - y_B)$$

Complexity : $O(Bd)$

$L(\theta) = \frac{1}{2n}\sum_{i=1}^{n}(\hat{y}^{(i)} - y^{(i)})^2 \leftarrow$ RSS $\quad \nabla_\theta L(\theta) = \frac{1}{n} X^T(\hat{y} - y)$

$L^{(i)}(\theta) = (\hat{y}^{(i)} - y^{(i)})^2 \quad \nabla_\theta L^{(i)}(\theta) = (\hat{y}^{(i)} - y^{(i)})\, x^{(i)}$

Formal implementation of Mini-batch

- ~~Initizi~~ Initialize θ
- For each epoch
  a) shuffle data
  b) partition data into mini-batches
  $B_1, B_2 \ldots B_{\lceil \frac{n}{B} \rceil}$, each of size B
  c) For each mini-batch
  $$\theta \leftarrow \theta - \frac{\eta}{B} X_{B_k}^T (X_{B_k}\theta - y_{B_k})$$
- Repeat untill ...